

Aryan Vijay Bhosale

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Education

2025 - 2027	University of Maryland, College Park <i>Master of Science in Computer Science</i> Advised by Prof. Dinesh Manocha & Prof. Ramani Duraiswami	College Park, MD, USA
2021 - 2025	Indian Institute of Technology, Bombay <i>Bachelor of Technology with Honours in Mechanical Engineering</i> <i>Minor in Artificial Intelligence and Data Science, Minor in Computer Science</i> Advised by Prof. Preethi Jyothi & Prof. Biplab Banerjee	Mumbai, India

Publications

- Aryan Vijay Bhosale et al. [VIBE: Video Instruction-aligned Background music gEneration](#), Accepted to **EMNLP 2026**
- Aryan Vijay Bhosale et al. [DuplexWorld: Live Evaluation Benchmark for Speech-to-Speech \(S2S\) Voice Agents](#)
- Aryan Vijay Bhosale et al. [TORUS: A Unified Benchmark for Unified Audio Models](#), Accepted to **EMNLP SALMA 2026**
- Lokegaonkar*, Bhosale* et al. [Video-Robin: Autoregressive Diffusion Planning for Video-to-Music Generation](#), Submitted to ECCV '26
- Belham*, Aryan Vijay Bhosale* et al. [Deep evolving semi-supervised anomaly detection](#), Submitted to ICLR '26
- Aryan Vijay Bhosale et al. [Anomaly detection using Diffusion-based method](#) *Equal contribution

Experience

Centific AI Research (CAIR) Redmond, WA, USA
Research Intern: Multimodal AI Group, Audio & Speech Group Jun '26 - Present

- Developed **DuplexWorld**, a live benchmark for speech-to-speech voice agents: GPT-Realtime, Gemini-Live, Grok Voice, Amazon Nova
- Developed **TORUS**, a unified benchmark testing generation-understanding self-coherence gap in unified audio models: NVIDIA Audex

Advisor: [Abhishek Mukherji](#), [Harshit Rajgarhia](#) | *Full-Duplex Speech, Speech Dialogue Models, Unified Audio Models, Live Evaluation Benchmarks*

Dolby Laboratories San Francisco, CA, USA
Research Collaborator: Dolby Advanced Technology Group (ATG) Aug '25 - Present

- Developed and trained **VIBE** & **Video-Robin** series of **SOTA** text+video-to-music generation models in collaboration with **NVIDIA**
- Achieved **2.21x** inference speed-up, outperforming SOTA models on audio quality, diversity & alignment metrics across **3** benchmarks

Advisor: [Lie Lu](#) | *Multimodal AI, Music Generation, Flow Matching, Diffusion Transformers, Distributed Training & Inference, PyTorch DDP*

American Express Gurugram, India
Data Science Intern: Model Interpretability (MI) Research May '24 - Jul '24

- Pioneered an **ICE-based** framework to quantify confidence in MI techniques (**SHAP, PDP, ALE**) on production credit risk models
- Benchmarked interpretability techniques on **XGBoost** credit (**6M+ data**) & **Gen. Linear Model** attrition (**4M+ data**) risk models

Advisor: [Aashish Prakash](#) | *Explainable AI, Statistical ML, XGBoost, statsmodels, SHAP, Scikit-Learn, Numpy*

Brown University Providence, RI, USA
Research Intern May '23 - Aug '23

- Implemented a **conditional Diffusion Model** for generating optimal mechanical structures, **8x** better performance than SOTA **GAN**
- Integrated mechanically guided sampling methods & gradient-free optimizers (**Genetic Algorithms, CMAES**) using **PyTorch** & **evosax**

Advisor: [Prof. Miguel Bessa](#) | *Generative AI, Deep Generative Models, Diffusion Models, DDPMs, PyTorch*

Technical Skills

Programming Languages	Python C++ C CUDA Bash/Linux LATEX Java SQL MATLAB R
Deep Learning	PyTorch TensorFlow Hugging Face (Diffusers, Transformers, PEFT, TRL) JAX
Generative AI	Stable Diffusion & Stable Audio LLaMa LoRA CLIP VAE LLaVa Open AI, Claude API
Vision & Audio	OpenCV Speechbrain Fairseq TorchAudio Torchvision Scikit-Image YOLO EasyOCR
ML Tools & Infrastructure	Git Docker Distributed Training (PyTorch DDP) Multi-GPU Training Hydra
Python Libraries	NumPy Matplotlib Scipy Pandas Scikit-Learn wandb

Achievements

- Received sponsorship of **INR 4M+** from **NVIDIA** for building India's first self-driving car with custom drive-by-wire & **ROS** software
- Awarded **Undergraduate Research Award** for exceptional deep learning research on Diffusion Models for continual anomaly detection

Service

Program Committee, NeurIPS Real-Time Conversational Agents (RTCA) Workshop 2026	Sep '26 - Dec '26
Reviewer, Association for Computational Linguistics (ACL) Rolling Review	Aug '26 - Sep '26
Reviewer, Indian Conference on Computer Vision, Graphics and Image Processing (ICVGIP) 2026	Sep '26 - Dec '26
Reviewer, ACM International Conference on Web Search and Data Mining (WSDM) 2026	Sep '25 - Oct '25
Research Assistant, Visual AI Lab, UK & Deep Visual Computing Lab, IIT Bombay	Dec '23 - Aug '25
Teaching Assistant, Computer Programming and Utilization Course (CS101), IIT Bombay	Jan '24 - May '24
Amazon Solve for Society Robotics Mentor	Aug '23 - Sep '23
Open Source Contributor, Hacktoberfest	Oct '23 - Nov '23
Academic Mentor, English Language Program Cabinet Member, Institute Web and Coding Convener, IIT Bombay	Jul '24 - Dec '24